

Jayden Fernandes

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PROFESSIONAL SUMMARY

Rutgers University School of Engineering student (Class of 2027) pursuing an Electrical Engineering internship in the automotive and EV industry. Hands-on experience in circuit simulation, PCB design, digital logic design, and embedded systems hardware integration. Proficient in LTspice, NGspice, KiCad, DesignSpark PCB, Qucs, and TinkerCAD Circuits; familiar with Python scripting and Linux-based environments.

EDUCATION

Rutgers University - New Brunswick

Bachelor of Science, School of Engineering.

Expected May 2027

Relevant Coursework: Calculus I/II/III, Differential Equations, Analytical Physics I & II, Digital Logic Design, Principles of Electrical Engineering I, Engineering Mechanics

Hopatcong High School

High School Diploma, Magna Cum Laude. GPA: 3.64 / 4.0 (Unweighted).

June 2023

EXPERIENCE

AI Task Creator and Executor

May 2026 – Present

Handshake AI

- Applied professional EDA tools including LTspice and NGspice to build and refine digital assets for a cutting-edge AI research project
- Designed and validated circuit simulation workflows across DesignSpark PCB, Qucs, and KiCad
- Produced structured schematic and PCB layout outputs to support AI training data pipelines

PROJECTS

Circuit Analysis & Simulation

Jan 2024 – May 2024

Principles of Electrical Engineering I Lab

- Simulated resistor-capacitor networks, voltage dividers, and transient circuits using LTspice
- Visualized signal waveforms and performed node voltage analysis to reinforce EE fundamentals
- Applied Kirchhoff's Laws, Thevenin equivalents, and time-domain analysis across circuit configurations

Digital Logic Circuit Design

Jan 2024 – May 2024

Rutgers University - Digital Logic Design Lab

- Designed and simulated combinational and sequential logic circuits using TinkerCAD Circuits
- Constructed virtual breadboard circuits incorporating logic gates, multiplexers, decoders, and flip-flops
- Applied Boolean algebra simplification techniques to optimize logic circuit implementations

Raspberry Pi GPS Speedometer

June 2022 – Aug 2022

Personal Project (Mentored)

- Collaborated with an IBM engineer mentor to build a GPS-based speedometer targeting an electric scooter navigation use case
- Integrated Navit, Raspberry Pi OS, and a USB GPS module to develop a functional hardware prototype
- Configured Linux environment and wrote Python scripts to interface hardware components

SKILLS

Engineering Concepts: Circuit analysis, Kirchhoff's Laws, Thevenin equivalents, Boolean logic, combinational/sequential circuits

Simulation & EDA Tools: LTspice, NGspice, KiCad, DesignSpark PCB, Qucs, TinkerCAD Circuits

Embedded Systems: Raspberry Pi, GPS navigation modules, Linux/Raspberry Pi OS

Programming: Python, Java (basic)

Languages: Fluent in Portuguese, Conversational in Spanish

VOLUNTEERING

- YMCA (Sept 2021 - May 2022): Organized and facilitated a youth yoga and jump rope program; monitored attendance and coordinated safe dismissal to parents.
- Color A Smile (July 2021 - July 2023): Created uplifting artwork distributed to military troops, nursing home residents, and community members.

ACTIVITIES

- OpenStreetMap (July 2022 - Present): Contribute to open-source mapping projects documenting roads, highways, and buildings in developing countries.
- Lego Robotics Club (Sept 2018 - Dec 2019): Designed and programmed Lego robots to execute directional commands.